# ARIADNE COLINA-VALERI

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## EDUCATION

Tufts University, Medford, MA — Expected Graduation Date: May 2028
Bachelor of Science in Electrical Engineering — GPA: 3.37/4.00
**Honors:** SHPE Scholar, HSF Scholar, 2024-2026 Dean's List, BEST Scholar, HIFF Scholar, Quest Bridge Scholar
**Relevant Coursework:** Junior Design, Embedded Systems, Circuits & Electronics, Linear Systems, Modern Physics

## PROFESSIONAL EXPERIENCE

**GE Transportation / Wabtec, Erie, PA** | *Control Systems Engineer Intern* — May 2026 – July 2026

- Automated locomotive diagnostic data extraction in Python, cutting a full-afternoon process to about an hour
- Built a Fault Analysis Module (FAM) from circuit schematics to flag failing parts across 63 locomotive devices
- Ran forced open & closed fault tests on each device, exposing an FPGA offset that skewed data differently per train

**Tufts Emerging Circuits and Systems Lab, Medford, MA** | *Research Assistant* — Sept. 2025 – April 2026

- Replaced a published RRAM memory model (CVAE) with a CTGAN to search for better-performing designs
- Grew the training data 10× to 5,000 samples & used a physics model (PINN) to reject impossible designs
- Found an Al2O3 design with 25% higher endurance & 11% lower switching energy than the published result

## SELECTED PROJECTS

**Linear Systems: UAV Pitch Control & RF Signal Restoration** — Jan. 2026 – May 2026

- Proved P & PI control can never stabilize the UAV, then grid-searched 15,625 PID gains to reach 2.6% overshoot
- Restored a blurred RF signal with a Wiener filter in MATLAB, cutting error by 77% compared with a direct inverse
- Showed a 2 s vs 0.5 s settling time makes RF recovery 1.5× worse, linking pitch control to signal quality

**Embedded Systems: DJ Board on an STM32 (Audio Recorder & Effects Processor)** — Jan. 2026 – May 2026

- Built a 4 kHz timer-driven state machine & removed audio noise, storing 5 s of 12-bit audio in 40 KB of SRAM
- Validated effect outputs (speed, pitch, etc.) via oscilloscope waveform analysis to confirm correct signal processing
- Implemented seven audio effects (echo, reverse, pitch, speed) enabling users to remix recorded sound in real time

**Digital Logic Circuits: FPGA-Based Audio Synthesizer** — Sept. 2025 – Dec. 2025

- Used a tuner to find notes playing an octave low from a clock at 2× the coded rate & rebuilt the 12-note scale
- Cut FPGA resource use from over 250% to within capacity by replacing division with bit shifts
- Fixed chord distortion with PWM amplitude mixing, enabling 6 clear notes across 8 octaves (16Hz–8kHz)

**Junior Design: Autonomous ESP32 Robot** — Sept. 2026 – Present

- Building a 7-state ESP32 state machine that authenticates websocket commands with a team key, rejecting all others
- Programming noise-filtered color & IR sensing that steers the robot to stay in its lane & avoid obstacles
- Designing a SolidWorks ladybug body & an ESP32 PCB with datasheet-chosen parts to replace breadboards

## ACTIVITIES & LEADERSHIP

**Residential Life & Learning** | *Resident Assistant* — Jan. 2026 – Present

- Respond to incidents & mediate disputes in high-pressure situations for a house of 60 residents

**IEEE / SHPE** | *Head Treasurer* — Sept. 2024 – Present

- Manage purchasing & annual budgets through Tufts Treasury, negotiating a 10% funding increase & preventing cuts

## TECHNICAL SKILLS

- **Programming:** C, C++, Python (PyTorch, pandas, CUDA), MATLAB, SystemVerilog, Arduino, UART, SPI, I2C
- **Hardware & Design:** AutoCAD, SolidWorks, Onshape, LTspice
- **Lab Tools & Components:** Oscilloscope, Analog Discovery 2 (logic/spectrum analyzer), FPGA, STM32, ESP32

## LANGUAGES & INTERESTS

- **Fluent:** English, Spanish, French; Beginner: Italian
- **Interests:** Singing, Rock Climbing, Artistic Swimming (21' USA FL JR. Champion; 22' Olympic Trials Qualifier)